Bartlett adjustment for random effects meta-analysis of event outcomes with small samples

Haben Michael[1]

[1]University of Massachusetts, Amherst, MA (hmichael@math.umass.edu)

Abstract. ((stijnen)) describes 4 random effects models for meta-analyses of involving event outcomes. Inference for these models is based on an asymptotic chi-squared approximation. We describe a procedure for Bartlett correction of the confidence interval, improving the convergence rate from $1/n$ to $1/n^2$ under ideal conditions. We apply the proposed procedure to meta-analyses considered by ((stijnen 2010)) on anti-infection devices, finding a differences in CI lengths up to 10%.

Keywords: Meta-analysis, Bartlett correction, Generalized linear mixed models

1 Introduction

There is currently underway a move to favor generalized linear mixed models in carrying out meta-analyses of event outcomes, particularly when sample sizes are small or events are rare. GLMMs avoid many of the dubious assumptions and devices, such as perfect knowledge of the study standard deviations and zero-cell corrections, required by the traditional alternative, inverse-variance weighting. However, inferences under GLMMs are still based on an asymptotic chi-squared approximation. Therefore, as in the case of inverse-variance weighted estimation, the conclusions of smaller meta-analyses are suspect, though smaller meta-analyses are where the use of GLMMs is often recommended. A classical remedy is Bartlett correction, which leverages modelling assumptions to improve the efficiency of the testing procedure. When the modelling assumptions are met, the improvement is typically a jump in the rate of asymptotic convergence from $1/n$ to $1/n^2$. [[the key modelling assumption in this setting is the gaussianity of the random effect.]]

While Bartlett corrections have long been applied to generalized linear models [[cite]], there do not appear to be extensions to even the simple GLMMs that are used in meta-analysis.[[verify this]] The reason for the reluctance appears to be computational difficulty [[cite hanada]]. While computer algebra systems have long been applied to compute the third- and fourth-order derivative arrays needed for Bartlett correction of GLMs ((cite old cordeiro paper, andrews and stafford, mccullagh maybe)), closed-form expressions for GLMM Bartlett corrections are unavailable for any of the commonly used models other than the

normal-normal/identity link ((cite noma, michael2026)). A combination of automatic symbolic manipulation and numerical approximation appears necessary. We argue that with proper design these difficulties can be overcome and consumer-grade hardware is sufficient for practical implementation.

In the next section we present four [[make consistent 4 vs four]] event data meta-analysis models, following [[Stifjnen's]] exposition. We then summarize the method of Bartlett correction, and discuss some of the underlying assumptions. In Section 3 we offer a practical approach to estimation of the Bartlett correction applicfable to the four models described in Section 2. In Section 4 we examine by simulation the effectiveness and robustness of the corrected test, finding that the proposed approximations are suitable for achieving the desired rate boost and identifying common model violations that may neutralize the advantage. In Section 5 we re-visit the data analyses of ((stijnen)). We find that, though the advantage of Bartlett correction may be negligible for moderate or large meta-analyses, it may be clinically relevant in smaller meta-analyses.

2 Background

2.1 Statistical models

We consider the random effects model for a meta-analysis consisting of n studies. In this model, a Gaussian random variable models the effects $\theta_1, \ldots, \theta_n$, of the individual studies in the meta-analysis,

$$\theta_i = \mu_0 + \tau_0 Z_i, \qquad Z_i \sim N(0,1)$$

The target of inference in the meta-analysis is μ_0, with τ_0^2 a nuisance parameter. The distribution of the data given the random effect depends on the type of analysis being conducted. Following ((stijnen)) we consider four types of analyses, determined by whether the studies included are single-arm or two-arm trials, and whether the estimand is based on proportions or incidence.

1. Single arm, proportions. Each study reports the number of events Y_i and total number of trials m_i. The statistical model is binomial

$$f_{m_i}(y_i \mid Z_i) = \binom{m_i}{y_i} p^{y_i}(1-p)^{m_i - y_i}$$
$$p = p(\mu_0, \tau_0^2) = \text{expit}(\mu_0 + \tau_0 Z_i)$$

   The target of inference μ_0 represents the log-odds of an event. The number of trials m_i is treated as a parameter of the conditional PMF f_{m_i}, which therefore typically varies across the studies in the meta-analysis.

2. Two arms, proportions. Each study reports, for treatment and control arm, the number of events $Y_i^{(T)}$ and $Y_i^{(C)}$ and the total number of trials $m_i^{(T)}$ and $m_i^{(C)}$. The statistical

model is conditional on the total number of events $S_i = Y_i^{(T)} + Y_i^{(C)}$, under which $Y_i^{(T)}$ follows Fisher's non-central hypergeometric distribution

$$f_{m_i^{(T)},m_i^{(C)},S_i}(y_i^{(T)} \mid Z_i) = \frac{\binom{m_i^{(T)}}{y_i^{(T)}}\binom{m_i^{(C)}}{S_i - y_i^{(T)}}\omega^{y_i^{(T)}}}{\sum_{\max(0,S_i-m_i^{(C)})}^{\min(S_i,m_i^{(T)})}\binom{m_i^{(T)}}{y}\binom{m_i^{(C)}}{S_i-y}\omega^{y}}$$

$$\omega = \omega(\mu_0, \tau_0^2) = \exp(\mu_0 + \tau_0 Z_i)$$

The target of inference μ_0 represents the ratio of the treatment to control log-odds. The parameters of the conditional PMF are $m_i^{(T)}, m_i^{(C)}$, and S_i.

3. Single arm, incidence rate. Each study reports the number of events Y_i and the duration of observation T_i. The model is Poisson,

$$f_{T_i}(y_i \mid Z_i) = \exp(-\lambda)\frac{\lambda^{y_i}}{y_i!}$$

$$\lambda = \lambda(\mu_0, \tau_0^2) = T_i \exp(\mu_0 + \tau_0 Z_i)$$

The target of inference μ_0 represents the log of the incidence rate. The follow-up time T_i is treated as a parameter of the conditional PMF.

4. Two arms, incidence rates. Each study reports, for treatment and control arm, the number of events $Y_i^{(T)}$ and $Y_i^{(C)}$ and the duration of observation $T_i^{(T)}$ and $T_i^{(C)}$. Conditionally on $S_i = Y_i^{(T)} + Y_i^{(C)}$, $Y_i^{(T)}$ is binomial

$$f_{T_i^{(T)},T_i^{(C)},S_i}(y_i^{(T)} \mid Z_i) = \binom{S_i}{y_i^{(T)}} p^{y_i^{(T)}}(1-p)^{S_i - y_i^{(T)}}$$

$$p = p(\mu_0, \tau_0^2) = \text{expit}(\log(T_i^{(T)}/T_i^{(C)}) + \mu_0 + \tau_0 Z_i)$$

The target of inference is the logarithm of the ratio of the incidence rates between the two arms. The sum of events S_i and the periods $T_i^{(T)}$ and $T_i^{(C)}$ are teated as parameters of the conditional PMF.

In all cases the likelihood for study i has the form

$$L_i(\mu, \tau^2 | Y_i) = \int_{-\infty}^{\infty} f_i(Y_i|z)\phi(z)dz, \qquad i = 1, \ldots n$$

where $f_i(y|z)$ is study i's PMF conditional on a Gaussian random effect, given above. Furthermore, the model parameters (μ, τ^2) only enter the likelihood ((ref)) through a parameter of the conditional PMF, the mean λ in the case of the Poisson, the success probability p in the cases of the binomial, or the odds ratio ω in the case of the non-central hypergeometric.

The likelihood ratio test of the null $\mu = \mu_0$ is based on the test statistic

$$W = 2\sum_{i=1}^{n}(\log L_i(\hat{\mu}, \hat{\tau}^2|Y_i) - \log L_i(\mu_0, \hat{\tau}_0^2|Y_i))$$

where $(\hat{\mu}, \hat{\tau}^2)$ maximizes $(\mu, \tau^2) \mapsto \sum_{i=1}^n \log L_i(\mu, \tau^2|Y_i)$ and $\hat{\tau}_0^2 = \hat{\tau}_0^2(\mu)$ maximizes $\tau^2 \mapsto \sum_{i=1}^n \log L_i(\mu_0, \tau^2|Y_i)$. In routine applications W converges to a chi-squared distribution under the null hypothesis, by Wilks' Theorem. Relying on this asymptotic approximation a confidence interval for μ_0 can be formed by inverting the test statistic.

2.2 Bartlett correction

Under a range of conditions [[citations]] the mean of a correctly specified LR statistic ((ref)) admits an expansion in integral powers of $1/n$

$$\mathbb{E}(W) = 1 + b/n + O(1/n^2).$$

Therefore, the Bartlett-corrected statistic

$$W_B = \frac{W}{1 + b/n}$$

has a mean agreeing with that of a chi-squared random variable up to order $O(1/n^2)$. The work of ((lawley and bartlett)) shows further that, under appropriate conditions, all other moments, and so also the entire distribution, also agree with a chi-squared up to order $O(1/n^2)$. This rate improvement holds even when the MLEs are plugged in for any unknown parameters appearing in the bias term b of the expansion ((ref)). As with ((ref lr statistic)), the adjusted statistic may be inverted to obtain CIs and p-values.

lawley ((cite)) provides a formula for the correction ((ref)) as sums and products of the components of the inverse Fisher information for ((ref likelihood)) and certain tensors [[k terms]]. Being a small set of sums and products this formula does not introduce numerical or computational difficulties that aren't already present in the basic tensor components input into the formula, and we focus here on the inputs. To this end, let D_r denote partial differentiation with respect to a sequence of the parameters μ and τ^2 given by r. For example, $D_{\mu,\tau^2} \sum_{i=1}^n \log L_i(\mu, \tau^2 \mid Y_i)$ denotes the off-diagonal term of the observed information matrix for ((ref total likelihood)). Define $k_r = k_r^{(0)} = \mathbb{E} D_r \sum_{i=1}^n \log L_i(\mu, \tau^2 \mid Y_i)$ and $k_r^{(s)} = D_s k_r$. In this notation, for example, the Fisher information matrix has components $k_{\mu,\mu}, k_{\mu,\tau^2}, k_{\tau^2,\mu}, k_{\tau^2,\tau^2}$. Lawley's formula for the Bartlett correction requires all terms k_r where r is of length 2, 3, or 4, and all terms $k_r^{(s)}$ where r is of length 3 and s of length 1 or r is of length 2 and s of length 1 or 2. Taking into account permutation symmetry of the order of differentiation, there are 35 such terms in a model with one parameter of interest and one nuisance paramter.

By linearity of expectation and differentiation, it suffices to compute the terms $k_r^{(s)}$ for the unit log-likelihoods $\log L_i(\mu, \tau^2 \mid Y_i), i = 1, \ldots, n$. Let us consider a single study from the meta-analysis, dropping the subscript i from the conditional PMFs and likelihoods. Then the general tensor component needed for the computation of Bartlett's correction is

$$k_r^{(s)} = \sum_y D_s \left(\left(D_r \log \int f(y|z)\phi(z)dz \right) \int w_k f(y|z)\phi(z)dz \right)$$

The outer sum is taken over the support of Y.

2.3 Assumptions

Among the assumptions underlying the above development we focus on three prominent ones. The first two stem from assumptions of the basic LR tests described in ((stijnen)), while the third is specific to Bartlett correction.

1. The first is correct specification of the likelihood, in particular, gaussianity of the unobserved random effect. [[we refer to sitjnen to other modeling assumptions implicit in ((ref 1–4 above)), such as ... .]] While the LR test itself depends on gaussianity for valid inference ((citation showing effects of bad estimatio of tau2)), the Bartlett correction may be less robust to deviations.

2. Wilks' theorem, underlying the chi-squared approximation to the LR statistic ((ref)), applies to points in the interior of the parameter space, excluding the fixed effects model $\tau_0^2 = 0$. It is possible the coverage rate at a given meta-analysis sample size n looks unaffected, as has been suggested ((cite)), while the more delicate rate analysis carried out here betrays an impact. [[this here or maybe in conclusion? conclusion is to be about relations bw fixed and random effects, and relation between LR and bartlett LR]]. with iv method can think of re model as a generalization of the fe, if the data is actually fixed effects CIs are valid there is just a loss of power. in the case of the lr, the random effects procedure is different from the fixed effects procedure.]]

3. the arguments under which the Bartlett correction delivers a $1/n^2$ rate are usually justified by Cramér's condition for the underlying random variables. This condition excludes lattice variables, such as are considered here. Consequently, while Bartlett corrections have been derived for statistics based on lattice random variables since at least the 80s, their effectiveness does not rest on a firm theoretical foundation and in fact does not appear to be understood in much generality ((cite frydenberg re multinomial ngative result and gotze, and janson re positive result)).

We have no theoretical contibutions to make here and will examine model robustness and the effect of discreteness by simulation.

3 Computing the tensor components ((k))

3.1 Overview

To compute the tensor component ((ref)), we approximate the integrals by a quadrature rule. Since all the likelihoods ((ref)) involve integrating against a Gaussian, Gauss-Hermite quadrature is a natural choice. This and similar methods ((cite biometrics paper)) provide weights $w_1, \ldots, w_{n_{GH}}$ and nodes $z_1, \ldots, z_{n_{GH}}$ in the domain of integration allowing the Gaussian integral of smooth integrands such as appear in ((ref)) to be approximated by a weighted sum,

$$k_r^{(s)} \approx \sum_y D_s \left(\left(D_r \log \sum_j w_j f(y|z_j) \right) \sum_k w_k f(y|z_k) \right)$$

As in ((ref exact version)), the outer sum is taken over the support of Y, which is a finite set in all the models considered here other than the Poisson model. The Poisson model requires a truncation point for computation, which introduces a tuning parameter into the proposed method. The number of nodes n_{GH} controls the quality of the approximation and introduces a further tuning parameter.

Carrying the differentiations ((ref above display)) through yields rational expressions in derivatives of the weighted sums $\sum_j w_j f(y|z_j)$, say, $F_{kl}(y) = \sum_j w_j \frac{d^k}{d\mu^k}\frac{d^l}{d(\tau^2)^l} f(y|z_j)$. These expressions for ((k)) in terms of the F_{kl} are the same for any model based on a likelihood of the form ((ref)) and may be pre-computed. For example, $k_{\mu,\mu} = \sum_y (F_{20}(y) - (F_{10}(y)^2/F_{00}(y)))$, and $k_{\mu,\tau^2}^{(\tau^2)} = \sum_y (F_{01}(y)^2 F_{10}(y)/F_{00}(y)^2 - (F_{02}(y)F_{10}(y) + F_{01}(y)F_{11}(y))/F_{00}(y) + F_{12}(y))$.

The derivatives $F_{jk}(y)$ of the model-specific conditional PMFs that appear in the 35 tensor components number 15: $F_{jk}(y)$ for $0 \le j + k \le 4$. The form of these derivatives may be derived independently of data but their specific values in general depend on parameters of the conditional PMF ((see above)), and must typically be computed for each study of a meta-analysis. Furthermore, since the conditional PMFs only depend on the model parameters (μ, τ^2) through the PMF's distributional parameters, the derivatives $F_{jk}(y), 0 \le j + k \le 4$, may be re-written as linear combinations [[linear for binomial and hyperg also??]] of $F_{00}(y) = \sum_j w_j f(y|\theta_j)$ using standard recurrences ((cite johnson and kotz univariate discrete distributions)). The advantage of this form is that the terms F_{00} involve only the PMF itself, for which optimized and numerically stable routines are available. For example, two applications of the Poisson PMF recurrence $\frac{d}{d\lambda} f_\lambda(y \mid Z) = f_\lambda(y-1 \mid Z) - f_\lambda(y \mid Z), y \ge 1$, yield, for model ((ref poisson/incidence rate model)),

$$\begin{aligned} F_{11}(y) &= \sum_j w_j \frac{d^2}{d\mu d\tau^2} f_{\lambda(\mu,\tau^2)}(y \mid z_j) \\ &= \frac{d\lambda}{d\mu}\frac{d\lambda}{d\tau^2} F_{00}(y-2) + (\frac{d^2\lambda}{d\mu d\tau^2} - 2\frac{d\lambda}{d\mu}\frac{d\lambda}{d\tau^2}) F_{00}(y-1) + \frac{d\lambda}{d\mu}\frac{d\lambda}{d\tau^2} F_{00}(y), \qquad y \ge 2 \end{aligned}$$

In summary. the approach is, for each of the n studies in the meta-analysis: for each y in the effective support of study i evaluate the 15 terms $F_{jk}(y)$ [[these are vectors across y actually]] using only routines for the conditional PMF (Poisson, binomial, or hypergeometric) and basic arithmetic, and then compute the ((35)) k terms using only the computed 15 ((F)) terms and basic arithmetic. The 15 $F_{jk}(y)$ terms can be computed in parallel, and once they are computed the 35 ((k)) terms can be computed in parallel. As the studies are independent the entire computation of the study-level Bartlett correction can be parallelized across the n studies, as well.

3.2 Tuning parameters

1. The main tuning parametr of the proposed method is n_{GH}, the number of nodes in the quadrature aproximation. check CIs for convergence. can be aided by visual inspection of the profile curves. visual examination recommended for any profile CI anyway to ensure the curve is approximately convex, [[other reasons]]. cite example in data analysis.

2. fixed vs adaptive. A further setting is whether fixed or adaptive GH is to be used. fixed GH uses the same choice of nodes irrespective of the integrand, namely, the roots of ..., which are centered at. on the other hand, adaptive GH transforms the nodes to take into account of the mode and scale of the integrand. difference in performance: poisson case, or other situations. where support points are far from 0. fixed nodes only grow

at root-n rate so very many are needed to capture the effect of those support points. computational tradeoff: adaptive requrise the mode and scale of the integrand must be recomputed on each call, but typically requires fewer nodes ((cite table)). furthermore, since the rule depends on the parametes mu,tau2 thorugh the mode and scale, the derivatives in ((ref k formula)) hit the weights and nodes as well, and become much more complicated. [[actually incompatible with method derived above]] intremediate approach, adopted above: we ignore this dependence, leading to an approximation intermediate in quality beween fixed and adaptive GH. this means that using adaptive over fixed GH is advantageous only when the mode and scfale of hte likelihood doesnt change to much under differentiation. when we do so, however, the cost of adaptive GH becomes very cheap. the only integrand for which adaptive GH needs to be computed are of the form condpmf phi. as all the cond pmfs belong to the exponential family, these integrands are strictly log-concave and unimodal, and the mode maybe obtained efficiently using Newton's method.

integrations applied to moments of variable, including exponential. premise of fixed rule nodes and weights are good enough for all functions and moments.

3. In the case of the Poisson-normal model, the number of support points to include. once a GH rule has been chosen, can choose the truncation point is large enough that the weighted tails are as small as desired. however it is difficult to know in general what is negligible for a given data set, and moreover the truncation point often influences the choice of GH rule. for the Poisson-normal model we therefore suggest examining a matrix of CIs varying both parameters to identify the limiting CI.

4 Simulation

We examine by simulation the rate at which the LR and Bartlett-corrected LR statistics converge to the reference chi-squared distribution under the four models ((ref)). The purpose is three-fold: 1. to investigate whether the quadrature approximations used by the proposed method are sufficient, 2. to investigate whether the Bartlett correction is effective as applied to this type of lattice data, as there are few theoretical results here, and 3. to assess the robustness the of any improvement in the rate offered by Bartlett correction to deviations from the assumed non-degenerate Gaussian model for the random effect. results are given in table ... and figure ...

4.1 Design

To make the simulation results relevant, the simulated meta-analyses are constructed to resemble the data of the ((n-w data sets, cite appendix)) analyzed in Section 5. The true effect μ_0 and heterogeneity τ_0^2 are taken to be the MLEs for the data obtained by ((stijnen et al.)), and the distributional parameters are obtained by sampling with replacement from the ((stijnen tables)). For example, for the control-arm ((bn-prop)) model, a simulated meta-analysis of size n is drawn by 1. sampling with replacement n trial sizes $m^{(C)}$ from the 18 control arm study sizes given by column ... of Table ...((ref nw18 data set)), 2.

sampling n effects Y under the binomial likelihood ((ref)) using the sampled trial sizes and the parameters $(\mu_0, \tau_0^2) = (...)$ taken from the MLE calculations of ((stijnen2010)). Meta-analyses of sizes $5, \ldots, 15$, are sampled in this manner, and the rate of convergence of the empirical Kolmogorov-Smirnov distance to a chi-squared distribution is estimated. ((see other ma paper for details on rate estimation)).

Robustness of the gaussianity assumption is examined by perturbing a normal to a skewed distribution ((cite pusey-jones paper)). the skew was chosen to test robustness since it drives the first-order error in CLT-type approximations. Robustness against non-degeneracy of the random effect, $\tau_0^2 > 0$, is studied in particular cases below.

In all simulations, we have used the true parameters (μ_0, τ_0^2) to calculate the Bartlett correction $b = b(\mu_0, \tau_0^2)$ [[perturbed by a Gaussian with SD $1/\sqrt{n}$??]]. In practice the MLE would be used. The practical reason is to avoid the prohibitive computational cost of re-calculating the Bartlett calculation for each of the n^4 monte carlo replicates ((this no. of mc reps wasnt actually mentioned yet)). The justification is the result of [[barndorff and cox]] that plugging in $\sqrt{n}-$consistent estimators into the Bartlett correction term $b(\mu, \tau^2)$, itself $O(1/n)$, only introduces an $O(1/n^2)$ error.

4.2 Results

table ... shows that when the assumptions on RE are met, the bartlett rate of 1/n2 appears to be achieved, with the excepton of the bn-irr model. lr rate also achieves 1/n rate. as skew increases, the rate deterioriates for both statistics. generally lr has a slower rate than the bartlett throughout, but for the largest skew tested and the bn-prop model, the bartlett corrected statistic shows a worse rate than the uncorrected. we therefore conclude that the quadrature approximations are sufficient and that skew affects the convergence of both corrected and uncorrected LR statistics.

The bn-irr model is the exception, showing a rate closer to 1/n than 1/n2 even in the idealized skew-free setting. this appears to be due to low heterogeneity. we vary the tau2 used to generate the data. rather than the mle (...) we examine tau2= ... . the results are presented in figure ... . the figure shows the 1/n2 rate being achieved as tau2 grows to ... . ((same for 1/n rate of LR stat?)) we therefore conclude that the advantage of Bartlett correction can be diminished by low heterogeneity.

5 Data analysis

We revisit the example data analyses of ((stijnen2010)). Meta-analyses are carried out under the models ((ref)) and 95% confidence intervals for the underlying effect μ_0 are calculated using the corrected and uncorrected profile LR. The results are presented in Table 3.

5.1 Proportions

((niel-wiese 2007)) carries out a meta-analysis on the efficacy of an anti-infection treatment for catheters. Each of the 18 included studies reports, for treatment and control arms, the total number of patients in the study and number of those who developed a catheter-related

Study	**Statistic**	0	0.05	0.1	0.15	0.2
bn.prop	LR (standard)	-1.14 (0.11)	-1.16 (0.14)	-0.90 (0.13)	-0.66 (0.11)	-0.08 (0.12)
	LR (Bartlett)	-2.12 (0.26)	-1.86 (0.25)	-1.05 (0.31)	-0.34 (0.32)	0.29 (0.20)
pn	LR (standard)	-1.16 (0.19)	-1.21 (0.17)	-0.69 (0.32)	-0.59 (0.03)	-0.52 (0.21)
	LR (Bartlett)	-1.89 (0.44)	-1.87 (0.34)	-0.96 (0.36)	-0.41 (0.02)	-0.29 (0.45)
bn.irr	LR (standard)	-1.16 (0.19)	-1.21 (0.17)	-0.69 (0.32)	-0.59 (0.03)	-0.52 (0.21)
	LR (Bartlett)	-1.89 (0.44)	-1.87 (0.34)	-0.96 (0.36)	-0.41 (0.02)	-0.29 (0.45)
hyperg	LR (standard)	-1.16 (0.19)	-1.21 (0.17)	-0.69 (0.32)	-0.59 (0.03)	-0.52 (0.21)
	LR (Bartlett)	-1.89 (0.44)	-1.87 (0.34)	-0.96 (0.36)	-0.41 (0.02)	-0.29 (0.45)

Table 1: Simulation results. The cells contain the medians and SDs of the log of the rate of the empirical Kolmogorov-Smirnov distance to a chi-squared distribution.

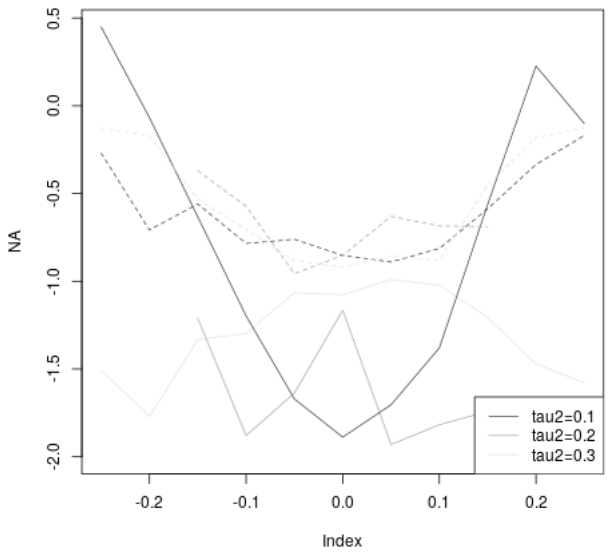


Figure 1: The effect of varying the heterogeneity parameter τ^2. The data is simulated based on ((nw2009)) and the binomial-normal model for incidence rate ratios.

infection ((ref Table. should mention this incorporates the corrections mentioned by metafor guy.)).

We apply the binomial-prpo model ((ref)) to obtain CIs for the log odds of infection in each arm separately and the hyperg model ((ref)) to obtain a CI for the log-odd ratio across the arms. In all three cases, the CI using the bartlett corrected profile likelihood is only slightly wider than the uncorrected, with the endpoints differing only in the hundredth's place or less. This agreement may reflect the moderate sample size of the the meta-analysis at 17–18 studies.[[effectively 17 for hn model due to double zero study]].

[[give sample plot]]

Adaptive		**Non-adaptive**	
n_{GH}	**Confidence Interval**	n_{GH}	**Confidence Interval**
20	[-3.8510, -3.1246]	50	[-4.1635, -2.8695]
40	[-4.0891, -2.9751]	150	[-4.0966, -2.9570]
60	[-4.0920, -2.9799]	250	[-4.0901, -2.9795]
80	[-4.0900, -2.9802]	350	[-4.0898, -2.9800]

Table 2: Comparison of the convergence of CIs using adaptive and fixed Gauss-Hermite rules. The meta-analysis data is the control arm of ((nw07)), analyzed under the single-arm binomial model ((ref)).

5.2 Incidence

niel-weise ((2008)) also carries out a meta-analysis on the efficiacy of an anti-infection treatment for catheters. Each of the 9 included studies reports, for each arm, the number of catheter-related infections observed in a population as well as the total "catheter-days", the sum over the population of the number of days each subject used a catheter.

We use the poisson-normal model ((ref)) to obtain CIs for the log incidence rate in each arm, and the binomial-irr model ((ref)) to obtain a CI for the log-incidence-rate ratio. For this smaller meta-analysis the difference between the corrected and uncorrected CIs is more dramatic. These differences include a loss of significance at the nominal level $\alpha = .05$ for the log-IR ratio study.

[[give sample plot. maybe give poisson model with adaptive and non-adaptive GH. cite latter in method/tuning parameter section.]]

[[update bn-prop trt analysis!!]]

6 Conclusion

[[maybe say postivie stuff here, below is negative]]

ideally we would applying the bartlett correction would be a free lunch. if the assumptions like gaussianity of the random effect are met, then we get the benefit of the rate improvement, and if they're not met, our inferences are the same as using the uncorrected statistic. we

Estimand	Meta-analysis	LR CI	Proposed CI	% Diff
log odds	stijnen-1, control arm	[-4.07, -3.00]	[-4.09, -2.98]	+3.7%
	stijnen-1, treatment arm	[-5.70, -4.20]	[-5.73, -4.19]	+2.7%
log-odds ratio	stijnen-1	[-2.21, -0.71]	[-2.25, -0.68]	+5.1%
log IR	stijnen-6, control arm	[0.90, 1.90]	[0.85, 1.95]	+10.6%
	stijnen-6, treatment arm	[0.10, 1.55]	[0.03, 1.62]	+9.8%
log-IR ratio	stijnen-6	[-1.05, -0.01]	[-1.12, 0.05]	+11.3%

Table 3: Standard and Bartlett-corrected confidence intervals for the example meta-analysis datasets considered by ((stijnen)). There is little advantage in Bartlett correction for the meta-analyses [[nw87 data]], which consists of 18 studies. There is a greater difference for the meta-analyses based on [[nw88 data]], which has half as many studies. [[switch first and second columns]]

	Integration Cutoff ($y_{\max}$)				
$n_{\mathbf{GH}}$	**100**	**200**	**300**	**400**	**1000**†
5	[0.9526, 1.7905]	[0.9585, 1.7592]	[0.9573, 1.7459]	[0.9568, 1.7407]	[0.9565, 1.7366]
6	[0.8713, 1.9132]	[0.8704, 1.9206]	[0.8684, 1.9236]	[0.8675, 1.9250]	[0.8669, 1.9282]
7	[0.8500, 1.9692]	[0.8525, 1.9580]	[0.8509, 1.9543]	[0.8501, 1.9529]	[0.8495, 1.9533]
8	[0.8557, 1.9431]	[0.8575, 1.9418]	[0.8558, 1.9409]	[0.8550, 1.9405]	[0.8444, 1.9418]
9	[0.8512, 1.9486]	[0.8531, 1.9471]	[0.8513, 1.9460]	[0.8505, 1.9456]	[0.8499, 1.9469]
10	[0.8496, 1.9512]	[0.8518, 1.9486]	[0.8500, 1.9473]	[0.8492, 1.9468]	[0.8486, 1.9479]
100†	[0.8495, 1.9503]	[0.8516, 1.9482]	[0.8499, 1.9470]	[0.8491, 1.9465]	**[0.8485, 1.9477]**

† Large parameter values serving as ground truth benchmarks.

Table 4: Convergence of Bartlett-corrected LR confidence intervals across Gauss-Hermite nodes (n_{GH}) and cutoff values ($y_{\max}$).

have seen this is not quite the case. when the skew of the random effect is sufficiently far from 0, the rate of the bartlett corrected sttaistic can be worse than the uncorrected statistic. Presumably a random effect sufficiently far from a Gaussian in other respects can also lead to this situation.

moreover, we have seen a similar qualification to the use of the basic LR statitsic,though it is not the focus of this paper. that is, it does not seem that the LR statistic is always better for RE inference than invese-variance weighting. applying the IV method when the the heterogeneity parameter τ_0^2 is in fact 0 does not by itself lead to an inconsistent test, though there may be a loss of power. however we have seen ((ref simulations)) that the convergence rate of the LR test (corrected or not) as the heterogeneity parameter approaches 0 can exceed 0, indicating inconsistency, at least as judged by the sample sizes considered here. This conclusion is however, consistent with our a priori expectation: [[maybe put self liang, nonorthogonality down here]]. ((however compare: ...give contrary results))